GM650 CYCLONE GROUND MOUNT KIT

Modular aluminium mounting system for JA Solar JAM72S42-LR (625–650 W) modules

Fabrication drawing package — for quotation and manufacture, Suva, Fiji

System summary

Module	2465 × 1134 × 30 mm, 29.6 kg, portrait
Tilt / facing	10° fixed, facing north, single row E–W
Bundles	1–12 panels: 1 × Starter Kit + n × Panel Kit
Wind design	70 m/s (252 km/h) ultimate gust, cyclonic
Material	Aluminium 6061-T6 (6063-T6 acceptable), SS A4 fasteners
Foundations	Concrete pads 500×500×600, 2 per trestle

Drawing index

PRELIMINARY — FOR QUOTATION AND PROTOTYPE. Indicative engineering to AS/NZS 1170.2 (70 m/s) and AS/NZS 1664.1. Obtain certification by a Fiji-registered structural engineer before commercial sale or installation. All dimensions in millimetres. Drawings not to scale — figured dimensions govern.

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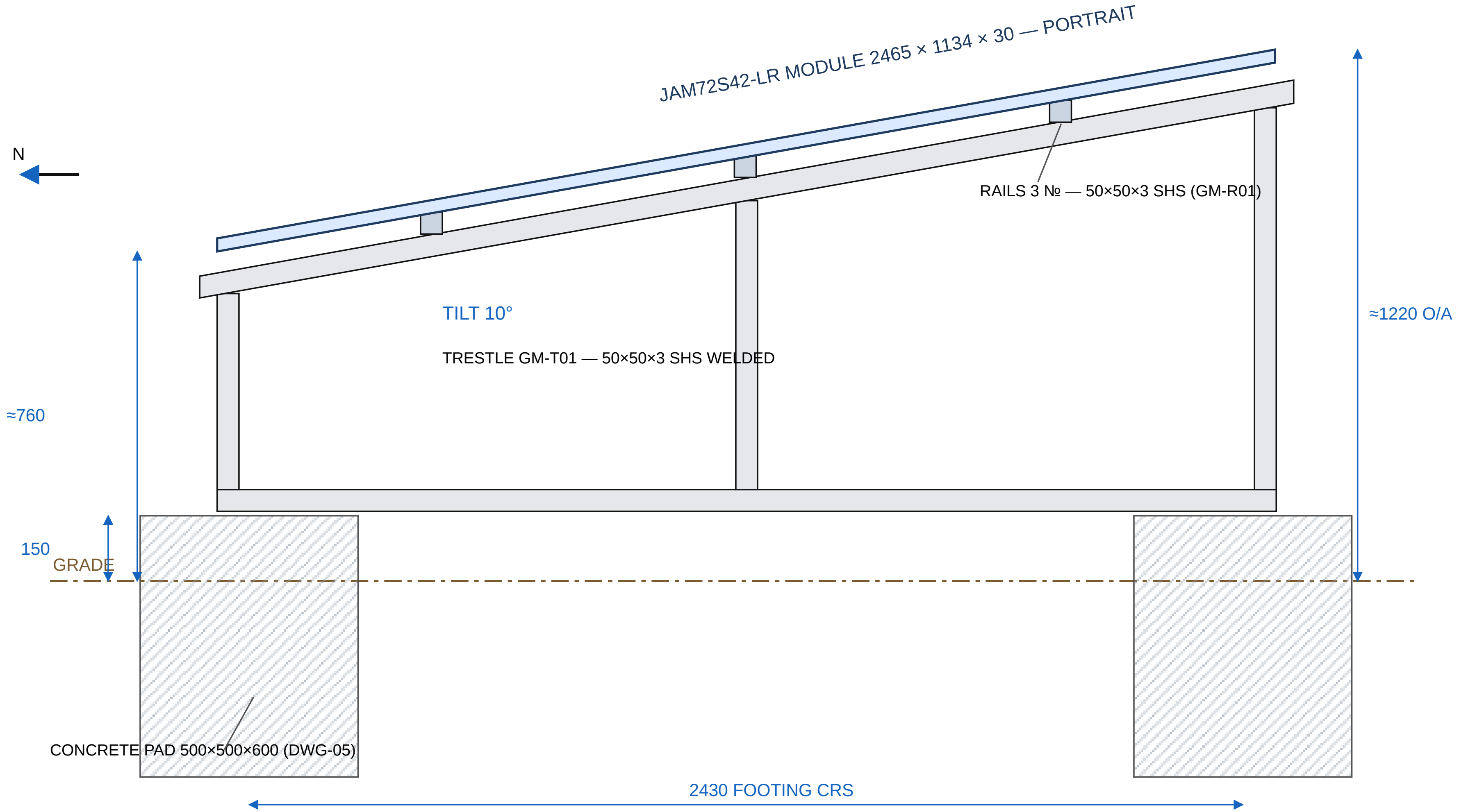
DWG-01 — GENERAL ARRANGEMENT (section through trestle, looking west)



Trestles at ≈1150 mm centres along row (n panels → n+1 trestles). Rails continuous over trestles, spliced with GM-S01 sleeves. Panel clamped to rails with imported 30 mm mid/end clamps — 6 clamp points per panel.

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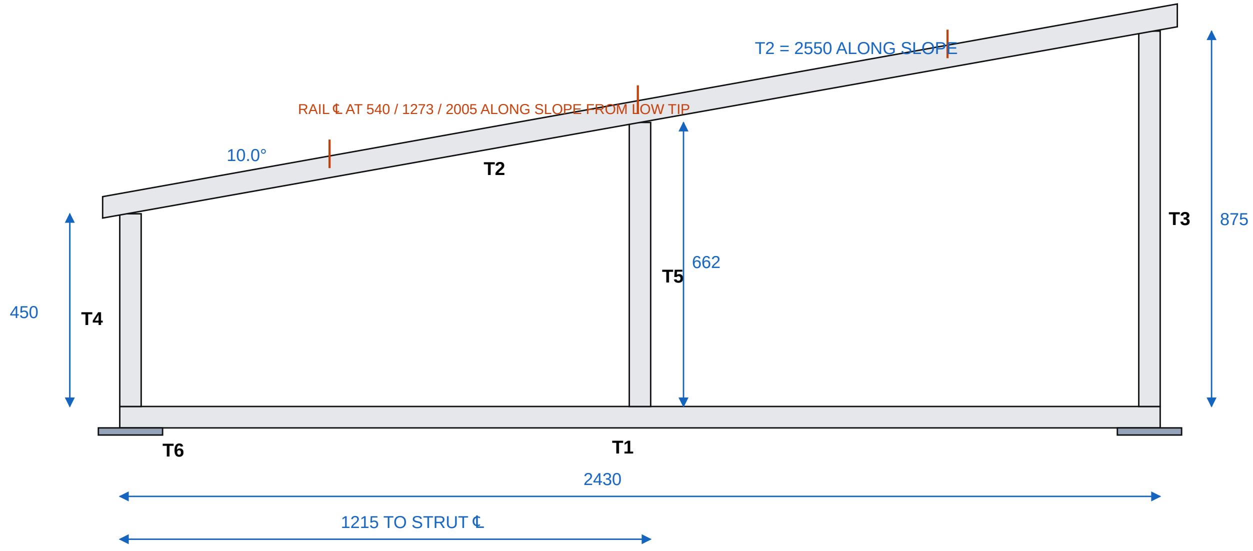
DWG-02 — GM-T01 TRESTLE FABRICATION (side elevation)



Cut list — one trestle (all 50×50×3 SHS, 6061-T6)

Mark	Member	Length	Ends
T1	Bottom chord	2430	square
T2	Rafter	2550	square
T3	Rear post	875	top 10°
T4	Front post	450	top 10°
T5	Mid strut	662	top 10°
T6	Baseplate ×2 — 150×150×8 plate, 2× Ø14 @ 100 crs, centred	weld under chord ends	

Fabrication notes

1. Posts T3/T4 flush with chord ends; strut T5 centred at 1215. Rafter bears on all three, low-tip overhang 40 beyond front post face.
2. Welds: 6 mm fillet all round every joint, filler 5356. Wire-brush clean.
3. Build in jig. Tolerance: overall ±3 mm, posts square to chord ±1°.
4. Baseplates parallel, hole pairs at 2430 crs frame-to-frame ±2 mm.
5. Mass ≈ 11.6 kg. Mill finish, no coating.

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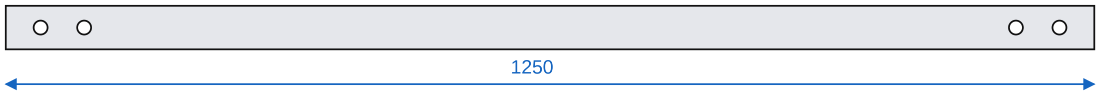
DWG-03 — GM-R01 RAIL, GM-S01 SLEEVE, SPLICE & CLAMP DETAILS

(a) GM-R01 RAIL SEGMENT — 50×50×3 SHS × 1250

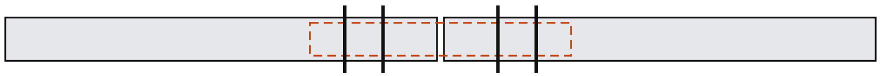


4× Ø9 THRU BOTH WALLS EACH END, AT 40 & 90 FROM END, ON ℄

(b) GM-S01 SPLICE SLEEVE — 40×40×3 SHS × 300

4× Ø9 THRU BOTH WALLS AT 40 & 90 FROM EACH END

(c) SPLICE ASSEMBLY

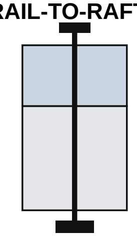


SLEEVE INSIDE, 150 EACH SIDE OF BUTT

4× M8×70 A4 HEX + NYLOC PER SPLICE

BUTT GAP ≤ 3. SPLICES ANYWHERE IN BAY; JOINT AT TRESTLE ALSO PERMITTED.

(d) RAIL-TO-RAFTER FIXING (at every trestle)

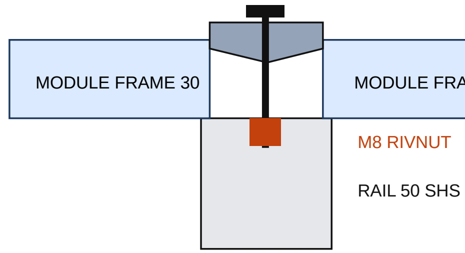


RAIL (50 SHS)

RAFTER (50 SHS)

M10×120 A4 + NYLOC — DRILL Ø11 VERTICAL
THRU BOTH ON SITE. TORQUE 30 Nm.

(e) CLAMP SECTION (imported)



MID CLAMP (30 mm) BETWEEN MODULES,
GAP 21. M8×25 A4 BUTTON HEAD +
SERRATED WASHER INTO M8 RIVNUT
SET IN RAIL TOP FACE (Ø11, SITE-SET).

END CLAMPS SAME FIXING AT ROW ENDS.
TORQUE 16 Nm (CONFIRM WITH CLAMP SUPPLIER).

CLAMP POSITIONS: ON EACH OF THE 3 RAILS —
RAILS SIT AT 0.2L / 0.5L / 0.8L OF MODULE:
WITHIN JA SOLAR APPROVED CLAMP ZONES.

IMPORTED PARTS: MID + END CLAMPS FOR
30 mm FRAME, BOLT-THROUGH TYPE (NOT T-BOLT),
+ M8 RIVNUTS. ALL ELSE LOCAL.

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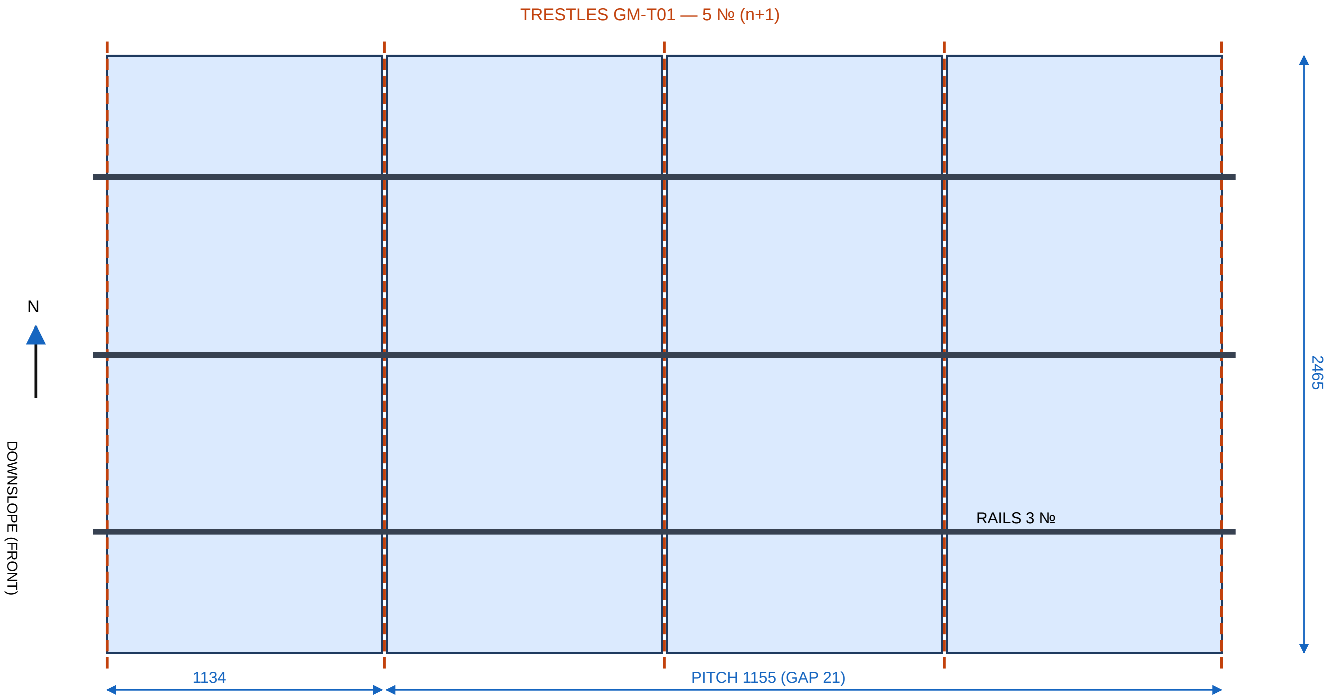
DWG-04 — ROW LAYOUT PLAN (4-panel bundle shown) & BUNDLE SCHEDULE



Bundle schedule (n panels = 1×SK + n×PK)

Bundle	SK	PK	BK	Trestles	Row length	Concrete
1	1	1	–	2	1.2 m	0.6 m³
2	1	2	–	3	2.3 m	0.9 m³
3	1	3	–	4	3.4 m	1.2 m³
4	1	4	–	5	4.6 m	1.5 m³
6	1	6	1	7	6.9 m	2.1 m³
8	1	8	1	9	9.2 m	2.7 m³
10	1	10	2	11	11.5 m	3.3 m³
12	1	12	2	13	13.9 m	3.9 m³

SK Starter Kit: 1 trestle + 6 end clamps + 1 bracing set + anchors.
PK Panel Kit: 1 trestle + 3 rails + 3 sleeves + 3 mid clamps + fasteners.
BK Bracing Kit: 2 straps + bolts, add 1 per 4 panels.
Rows > 12 panels: start a second row ≥ 4 m clear to the south.

Outer trestles under outer module edges; interior trestles at module joints (≈1150 crs). Trim final rail segments to 60 overhang past outer module edge.

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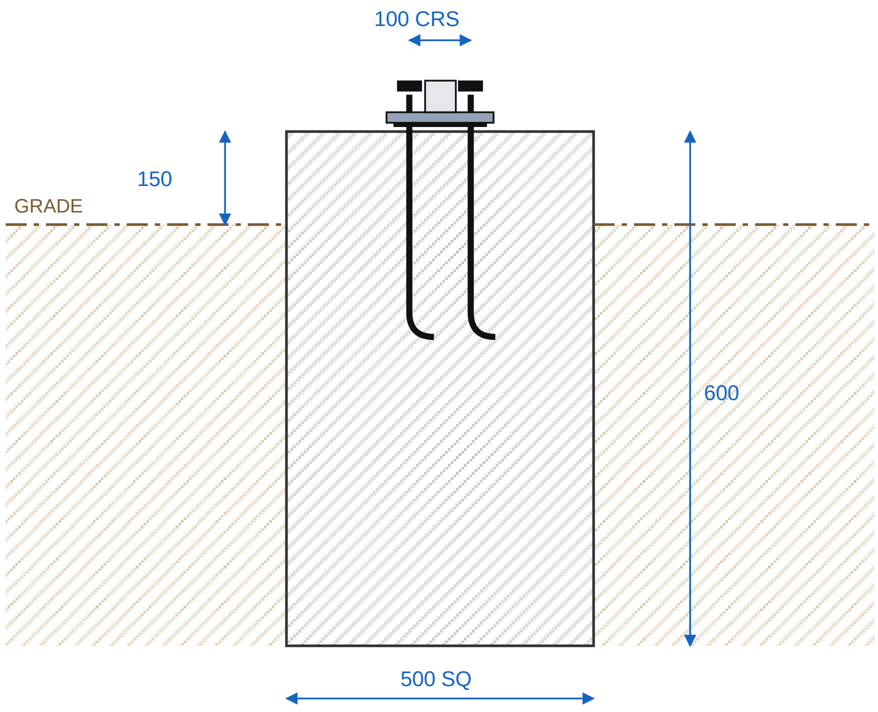
DWG-05 — FOOTING & ANCHORAGE DETAIL



Notes

1. Pad 500 × 500 × 600 deep, 20 MPa concrete, top 150 above grade, trowel level. Two pads per trestle at **2430 crs** (front–rear).
2. Anchors: 2 × M12 hot-dip-galv J-bolts per baseplate at 100 crs, projecting 60. Set with baseplate or ply template as jig while pouring. **Alternative:** 2 × M12 A4 chemical studs, 110 embed, drilled after 2-day cure.
3. 3 mm EPDM pad between baseplate and concrete (aluminium–concrete isolation).
4. Nut + flat washer, torque 45 Nm after row alignment.
5. Design uplift ≈ 5 kN ultimate per footing. In soft, filled or waterlogged ground increase to 600 × 600 × 700 and confirm with engineer.
6. No reinforcement required at this size; a single 665-mesh offcut mid-depth is cheap insurance where available.

Setting out

String both footing lines 2430 apart; anchor pairs positioned so trestle baseplate holes (Ø14) land centrally. Check diagonals across each trestle position before pouring.

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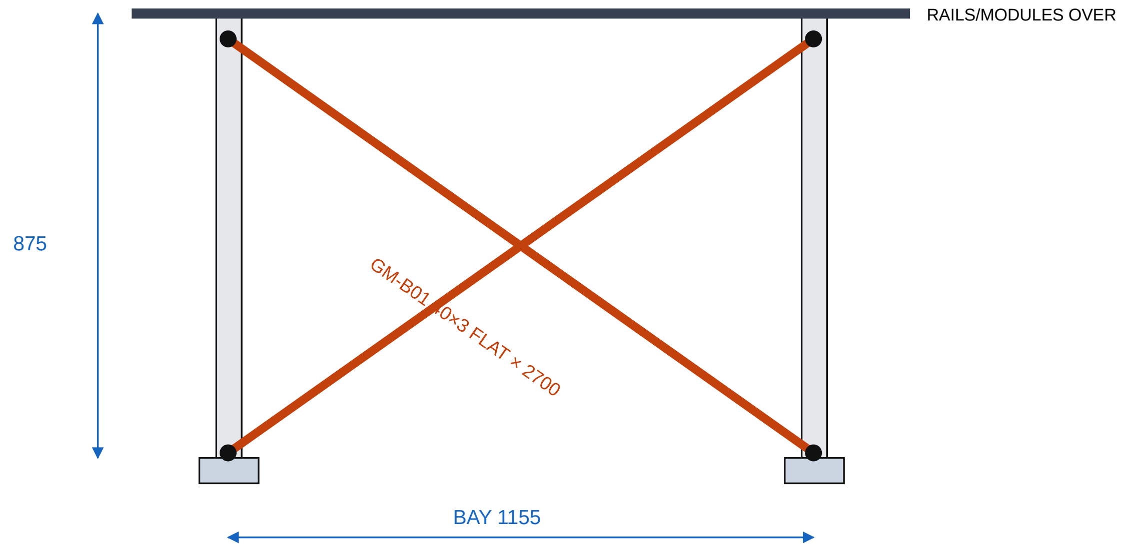
DWG-06 — GM-B01 ROW BRACING (elevation looking south, one braced bay)



REAR POSTS OF ADJACENT TRESTLES SHOWN; FIT MATCHING PAIR ON FRONT POSTS OF SAME BAY.

Notes

1. Crossed strap pairs resist along-row (E–W) wind. Fit in the **first bay, last bay, and every 4th bay** — on both the rear-post plane and front-post plane of the braced bay (4 straps per braced bay = 2 × GM-B01 sets).
2. Strap: 40×3 flat bar × 2700, Ø11 hole 20 from each end. Trim on site.
3. Fix M10×20 A4 + nyloc through post/chord walls, holes site-drilled Ø11. Pull straps hand-tight before final torque (20 Nm).
4. Bundles of 1–2 panels: single braced bay is sufficient (SK includes one set — fit rear plane; add front plane straps from the set).

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GENERAL NOTES, FASTENER SCHEDULE & IMPORTED ITEMS

General

1. All dimensions mm. Figured dimensions govern; do not scale drawings.
2. Aluminium: 6061-T6 preferred; 6063-T6 acceptable. Mill finish throughout.
3. Welding: 6 mm fillets, 5356 filler, structural-aluminium-experienced welder, jig-built frames, first article required.
4. Cut tolerance ±2, holes ±1, trestle geometry ±3 / 1°.
5. Design: 70 m/s ultimate gust (AS/NZS 1170.2 cyclonic basis), members to AS/NZS 1664.1. ≈10 kN ultimate uplift per interior trestle.
6. PRELIMINARY — certify with a Fiji-registered structural engineer before commercial sale. Module clamping to comply with JA Solar installation manual.
7. Panels face north at 10°; row runs east–west.

Stock per Panel Kit (from 6.0 m bars)

Stock	Consumption
50×50×3 SHS	≈1.5 bars (nesting: 3 bars → 2 trestles + 3 rails)
40×40×3 SHS	0.15 bar (20 sleeves/bar)
8 mm plate	2 × 150×150 baseplates
40×3 flat	SK/BK only — 2 straps per bar

Fastener schedule (stainless A4/316)

Fastener	SK	PK	Use / torque
M8×70 hex + nyloc + washers	–	12	Splices — 20 Nm
M8×25 button + serrated washer	6	3	Clamps — 16 Nm
M8 rivnut, large flange	6	3	Rail top face, site-set
M10×120 hex + nyloc	3	3	Rail–rafter — 30 Nm
M10×20 hex + nyloc	4	–	Bracing — 20 Nm (also 4/BK)
M12 J-bolt HDG (or A4 chem stud)	4	4	Anchors — 45 Nm
EPDM pad 150×150×3	2	2	Plate isolation
M8 earth lug + 6 mm^2 Cu	3	–	Per AS/NZS 5033

Imported items (only these)

Item	Spec	SK	PK
End clamp	30 mm frame, bolt-through M8 (not T-bolt)	6	–
Mid clamp	30 mm frame, 21 gap, bolt-through M8	–	3
Rivnut setter	M8 hand tool (one-time)	1 per install crew	

Everything else — sections, plate, stainless fasteners, anchors, EPDM, concrete — is available from Suva metal/hardware/marine suppliers.

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